

JEE MAIN REVISION MASTER: WAVE OPTICS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 26 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Diffraction, Polarization & Brewster's Law	14 PYQs	Diffraction & Polarization
High	Interference of Light & Young's Double Slit Experiment	12 PYQs	Interference

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Diffraction, Polarization & Brewster's Law	$\beta_0 = \frac{2\lambda D}{a}, \quad n = \tan i_p, \quad I = I_0 \cos^2 \theta$
Interference of Light & Young's Double Slit Experiment	$\beta = \frac{\lambda D}{d}, \quad \frac{I_{\max}}{I_{\min}} = \left(\frac{a_1 + a_2}{a_1 - a_2} \right)^2, \quad \beta \propto \lambda$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Diffraction, Polarization & Brewster's Law (High Yield — 14 PYQs)

Core Principle: Single slit Fraunhofer diffraction: Minima condition $a \sin \theta = n\lambda$. Width of central maximum $\beta_0 = (2\lambda D / a)$ (twice the width of secondary maxima). Brewster's Law for polarization by reflection: $n = \tan i_p$. At the Brewster angle, the reflected and refracted rays are mutually perpendicular ($i_p + r = 90^\circ$). Malus' Law: $I = I_0 \cos^2 \theta$.

■ **Caution / JEE Pitfall:** Diffraction occurs with sound and light, but POLARIZATION is unique to transverse waves (sound waves cannot be polarized!).

■ **Speed Trick:** When unpolarized light of intensity I_0 passes through an ideal Polaroid, the transmitted intensity is ALWAYS exactly $I_0/2$, regardless of orientation.

Representative Source PYQs:

- [2025 Shift] Q#2 (MCQ): "If A is incident plane wave front then refracted wave front in ($n_2 > n_1$)..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "Three polaroid sheets are co-axially placed as indicated in the diagram. Pass axes of the polaroids 2 and 3 make 30° and 90° ..." [Ans: B]
- [2023 Shift] Q#1 (MCQ): "When light propagates through a given homogeneous medium, the velocities of..." [Ans: N/A]

3.2 Interference of Light & Young's Double Slit Experiment (High Yield — 12 PYQs)

Core Principle: Conditions for interference: coherent sources (constant phase difference). Constructive interference: path difference $\Delta x = n\lambda$, intensity $I_{\text{max}} = (\sqrt{I_1} + \sqrt{I_2})^2$. Destructive interference: $\Delta x = (2n-1)(\lambda/2)$, $I_{\text{min}} = (\sqrt{I_1} - \sqrt{I_2})^2$. Fringe width $\beta = (\lambda D / d)$.

■ **Caution / JEE Pitfall:** If YDSE apparatus is immersed in water of refractive index n , fringe width decreases to $\beta' = \beta/n$ because $\lambda' = \lambda/n$.

■ **Speed Trick:** If one slit of YDSE is covered with a transparent sheet of thickness t and refractive index μ , the entire fringe pattern shifts by $y = (D/d)(\mu - 1)t$ without changing fringe width.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "In Young's double slit experiment, how many maxima can be seen on a screen (including central maxima) if $d = (5\lambda/2)$ (where λ is wavelength of light)." **[Ans: N/A]**
- **[2024 Shift] Q#4 (MCQ):** "In Young's double slit experiment, an electron beam is used to produce interference fringes of width β_1 . Now the electron beam is ..." **[Ans: N/A]**
- **[2023 Shift] Q#3 (MCQ):** "In the Young's double slit experiment, the intensity of light passing through each of the two double slits is $2 \times 10^{-2} \text{ W m}^{-2}$. Then..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

In Young's double slit experiment, how many maxima can be seen on a screen (including central maxima) if $d = (5\lambda/2)$ (where λ is wavelength of light and d is distance between the two slits).

- (A) 5
- (B) 4
- (C) 7
- (D) 1

Correct Answer: None

Solution Analysis: For bright fringes or maxima in Young's double slit experiment, the path difference must be an integral multiple of wavelength: $d \sin \theta = n\lambda$ where $n = 0, \pm 1, \pm 2, \dots$. Since the maximum possible value of $\sin \theta$ is 1, $|n\lambda| \leq d$. Given, $d = (5\lambda/2)$. So, $|n| \leq (d/\lambda)$ $|n| \leq (5/2)$ $|n| \leq 2.5$. But n must be an integer. So possible values of n are $n = -2, -1, 0, 1, 2$. That gives total maxima: 5. So the correct option is: 5 Option A.

Benchmark Problem 2: KCET 2024 (Q#3)

Three polaroid sheets are co-axially placed as indicated in the diagram. Pass axes of the polaroids 2 and 3 make 30° and 90° with pass axis of polaroid sheet 1. If I_0 is the intensity of the incident unpolarised light entering sheet 1, the intensity of the emergent light through sheet 3 is

- (A) zero
- (B) $\frac{I_0}{32}$ [Correct]
- (C) $\frac{I_0}{8}$
- (D) $\frac{I_0}{16}$

Correct Answer: B

Solution Analysis: For unpolarised light falling on the first polaroid, the transmitted intensity becomes half. So after sheet 1, $I_1 = \frac{I_0}{2}$. Now the pass axis of sheet 2 makes an angle 30° with sheet 1. Using Malus' law, $I_2 = I_1 \cos^2 30^\circ$. Since $\cos 30^\circ = \frac{\sqrt{3}}{2} \Rightarrow \cos^2 30^\circ = \frac{3}{4}$ we get $I_2 = \frac{I_0}{2} \cdot \frac{3}{4} = \frac{3I_0}{8}$. Now sheet 3 makes 90° with sheet 1, while sheet 2 makes 30° with sheet 1. So the angle between sheets 2 and 3 is $90^\circ - 30^\circ = 60^\circ$. Again us